

Marvin L. Gentry
Independent Researcher
Seattle, WA, USA

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The Editorial Board
Journal of Geometry and Physics
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Dear Editors,

I submit the manuscript “**Discrete Mixing Operators from Boundary Sector Geometry**” for consideration in the *Journal of Geometry and Physics*.

Summary. We develop a kinematic framework in which unitary mixing matrices arise as overlap operators between sector subspaces of a Hilbert space associated with a compact hyperbolic 3-manifold. Assuming a discrete assignment of sectors to boundary directions of the manifold (a geometric rather than dynamical input), we prove three structural theorems.

The *Geometric Dependence Theorem* (Theorem 5.1) establishes that the mixing matrix U_{ST} is determined entirely by the mutual positions of the sector subspaces, independently of any choice of basis within each sector.

The *Stability Theorem* (Theorem 6.1) shows that rephasing-invariant mixing observables — the Jarlskog invariant and all moduli $|U_{ij}|$ — are stable under bounded deformations of the sector configuration that preserve the discrete assignment rule.

The *Discreteness Theorem* (Theorem 6.2) establishes that the set of admissible mixing matrices is discrete in $U(n)$ modulo $U(1)^n \times U(1)^n$, so that no continuous deformation of physical parameters can produce continuous drift in any rephasing-invariant mixing observable.

The construction requires no dynamical input and provides a model-independent structural constraint on theories in which fermions are associated with boundary directions in a compactified hyperbolic geometry.

Fit with JGP. The paper sits at the intersection of hyperbolic geometry, operator theory, and particle physics that defines JGP’s scope. The primary contribution is mathematical (three proved theorems about geometric operators on hyperbolic manifolds); the connection to Standard Model mixing matrices provides physical motivation and application.

Prior submission. The manuscript was previously submitted to *Letters in Mathematical Physics* (MATH-D-26-00165), which declined it without substantive comment. It has not been submitted elsewhere and is not currently under review at any other journal. A preprint is available at SSRN (Abstract ID: 6689441, <https://papers.ssrn.com/abstract=6689441>).

Suggested reviewers. Given the geometric and representation-theoretic nature of the

results, I suggest the following potential reviewers: Alan Weinstein (UC Berkeley), Peter Michor (Vienna), Nigel Hitchin (Oxford).

Yours sincerely,

Marvin L. Gentry
Independent Researcher
Seattle, WA, USA
drmlgentry@protonmail.com
ORCID: 0009-0006-4550-2663